

Dong Payton Pei

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SUMMARY

Software engineer focused on AI agents and developer tools, with Rust/Tauri, TypeScript/React and Python experience. Built local-first agent applications and shipped LifeMint on the App Store. Former Nokia and Ford co-op; currently studying Systems Science and Engineering at the University of Ottawa. Seeking software engineering opportunities in Hangzhou.

EDUCATION

University of Ottawa

Master of Systems Science and Engineering

2024 — Present

Carleton University

Bachelor of Computer Science

2017 — 2023

PROJECTS

pipi-shrimp-agent

github.com/mammuto01/pipi-shrimp-agent

- Built a local-first desktop AI agent with Rust/Tauri, React/TypeScript and SQLite, integrating multiple LLM providers, structured tool calls and project memory.
- Implemented provider adapters, buffered SSE parsing and cancellation, plus CDP browser automation with failure recovery and context compression for long-running sessions.

Conveyor

github.com/mammuto01/Conveyor

- Built a self-hosted Python service connecting Telegram and Feishu to Codex CLI, separating code changes into detached Git worktrees with diff, apply, discard and cancellation controls.
- Implemented a persistent SQLite task queue with restart recovery, operator allowlists, explicit approval gates and audit logging.

LifeMint (formerly Focus Mint)

apps.apple.com/us/app/lifemint/id6759029810

- Published a SwiftUI productivity app for iPhone, iPad and Apple Watch with focus sessions, earnings tracking and English, Chinese and French localization.
- Implemented StoreKit purchases, widgets, Live Activities and bidirectional iPhone–Apple Watch synchronization with WatchConnectivity.

Resume Generator

github.com/mammuto01/Resume-Generator

- Built a React/TypeScript and Node.js resume workspace with PDF/text import, job-description tailoring and Typst PDF/SVG export.
- Added Zod request validation, Docker deployment and automated tests with Vitest and Playwright.

WORK EXPERIENCE

Software Tester Co-op

Aug 2020 — Aug 2021

Nokia Canada

- Enhanced JavaScript/XML code to improve code coverage and robustness.
- Produced Confluence design documentation for system architecture and data flows.

Software Developer Co-op

Apr 2020 — Aug 2020

Ford Motor Canada

- Participated in Agile development, wrote C++11 unit tests with GoogleTest, and used Jenkins in the CI/CD workflow.

RESEARCH

Low-Light RAW Image Restoration with Bridge U-Net and DDBM

2026

- Responsible for the initial DDBM implementation in a collaborative low-light RAW image restoration study using SID and ELD data.
- The project evaluated a DDBM-Feature variant at 26.12 dB PSNR and 0.807 MS-SSIM on 598 SID Sony test images.

SKILLS

- Languages:** Python, TypeScript, JavaScript, Rust, Swift, C++
- Frameworks:** React, Node.js, Tauri 2, SwiftUI
- AI & Systems:** LLM APIs, Tool Calling, MCP, SQLite, CDP Browser Automation, WatchConnectivity, StoreKit 2
- Testing & Delivery:** PyTest, Vitest, Playwright, GoogleTest, Jenkins, Docker